

Bohdan Palii

FULL-STACK ENGINEER

TypeScript, Node.js (NestJS), React, Next.js, React Native, PostgreSQL

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Summary

Full-Stack Engineer with 5 years of experience building production SaaS platforms for healthcare, fintech, logistics and education clients across the UK, EU and US. TypeScript across the stack: NestJS and Node.js REST APIs, React and Next.js web applications, React Native (Expo) mobile apps published to the App Store and Google Play, and PostgreSQL. Strong in multi-tenant architecture, asynchronous processing (Redis, BullMQ, RabbitMQ, Pub/Sub), payment integrations (Stripe, Paysera) and LLM/RAG features (OpenAI and Anthropic APIs, Vercel AI SDK, pgvector). Comfortable owning features end-to-end in distributed, English-speaking teams, with GDPR-compliant data handling.

Skills

Languages: TypeScript, JavaScript (ES6+), HTML5 / CSS3, SQL, Python (FastAPI), PHP

Frontend: React, Next.js (SSR), React Router, Redux, Redux Saga, Zustand, TanStack Query (React Query), RxJS, Tailwind CSS, SCSS, shadcn/ui, Radix UI, Material UI (MUI), Chakra UI, Bootstrap, Bulma, Storybook, Framer Motion, GSAP, Three.js / React Three Fiber, Vite, Webpack, PWA, i18n, Zod, jQuery, Electron.js

Mobile: React Native, Expo, Expo Router, push notifications, App Store and Google Play publishing

Backend: Node.js, NestJS, Express.js, Fastify, REST APIs, OpenAPI / Swagger, tRPC, WebSockets (Socket.IO), BullMQ, RabbitMQ, Google Cloud Pub/Sub, Cloud Tasks, JWT authentication, RBAC, multi-tenant architecture, event-driven architecture, background workers, Strapi, WordPress

Databases & ORMs: PostgreSQL, PostGIS, pgvector, MySQL, MongoDB, Redis, Elasticsearch, Firebase / Firestore, Supabase, Prisma, TypeORM, Drizzle ORM, Sequelize

Cloud & DevOps: AWS (S3, ECS, RDS, CloudFront), Google Cloud (Cloud Run, Cloud SQL, Pub/Sub, Cloud Functions), Vercel, Docker, Kubernetes, Nginx, MinIO, CI/CD, GitHub Actions, GitLab CI, Git / GitHub / GitLab, Sentry

AI & LLM: OpenAI API, Anthropic API, Vercel AI SDK, RAG pipelines, vector search (pgvector), Ollama, Groq API, Deepgram, AssemblyAI, GitHub Copilot, Claude Code, Codex, LLM evaluation, RLHF data generation, agent evaluation, rubric design, grading calibration, failure-mode analysis

Payments & Integrations: Stripe, Paysera, Cliniko, Nookal, Meddbase, Semble, Ticketmaster / AXS, ArcGIS, Mapbox, SCORM, EDI, Finvalda, Ivesk.It

Testing & Quality: Jest, Vitest, React Testing Library, Playwright, Puppeteer, ESLint, AES-256 encryption, Helmet, GDPR-compliant data handling

Education

Cherkasy State Business College

Bachelor of Science, Software Engineering · Cherkasy, Ukraine

2020 – 2024

Northern Virginia Community College

2023 – 2024

Languages

English – C1 (certified); Ukrainian – Native

Additional Information

Based in Alicante, Spain (CET). Open to on-site and hybrid work in Spain, remote work across the EU, and relocation.
Work authorization: Spanish residence and work card (TIE/NIE) scheduled for issue in October 2026.
I consent to the processing of my personal data for recruitment purposes in accordance with the GDPR (Regulation (EU) 2016/679).

Work Experience

UAPP LLC Full-Stack Developer · Remote – Wilmington, Delaware, USA	09/2025 – Present
CubeX Full-Stack Developer · Remote – Zaporizhzhia, Ukraine	07/2024 – 09/2025
ACCA Full-Stack Developer · Hybrid – Fairfax, Virginia, USA	10/2023 – 07/2024
WelSpot Inc. Full-Stack Developer · Remote – Miami, Florida, USA	09/2022 – 10/2023

Projects

Motics — Healthcare Multi-Agent Platform Full-Stack Developer / Software Architect · United Kingdom	• Led architecture and full-stack development of a healthcare SaaS platform from conception to production, now used by clinics across the United Kingdom. Built as an Nx/pnpm monorepo covering the Next.js web app, the React Native (Expo) mobile app and the backend services. • Expanded the product from an AI medical scribe into a clinic operations suite: clinical documentation, billing, patient communication automation and agent-driven workflows. • Designed asynchronous worker infrastructure on Google Cloud – Cloud Run, Pub/Sub and Cloud Tasks with Redis – to run AI workflows, speech-to-text transcription and media pipelines, with provider-neutral transcription across Deepgram, AssemblyAI and LLM providers. • Built the durable processing plane for the scribe: durable job projections into the web database, summary-source cutover, stranded-recording recovery and golden parity fixtures that hold prompt output stable across model changes. • Built and extended a dedicated FastAPI integrations service on Cloud Run that owns provider credentials, OAuth, webhook subscriptions and UK regional shard routing, with Alembic migrations, Secret Manager and Sentry/Amplitude observability.
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- Delivered EHR integrations for Cliniko, Nookal, Meddbase, Semble, Splose and Dentally, including clinical note push, identity mapping and reconciliation; plus Healthcode and Stripe billing, Gmail and Microsoft 365 mailbox sync, Twilio SMS, and Vald sports-science devices (ForceDecks, ForceFrame, NordBord, HumanTrak, SmartSpeed, DynaMo).
- Migrated production data from Firebase to PostgreSQL with Drizzle ORM, implemented Stripe billing, and published the mobile apps to the Apple App Store and Google Play with push notifications, Apple and Google sign-in, and crash reporting.

Stack: TypeScript, Next.js, React, NestJS, Node.js, React Native (Expo), Python, FastAPI, PostgreSQL, Drizzle ORM, Firebase / Firestore, Google Cloud (Cloud Run, Cloud SQL, Pub/Sub, Cloud Tasks, Secret Manager), Vercel AI SDK, OpenAI, Anthropic, Deepgram, AssemblyAI, Stripe, Twilio, Tailwind CSS, shadcn/ui, Radix UI, TanStack Query, Sentry, Amplitude, Playwright, Vitest, Docker

Agentic coding evaluation programme — Surge AI (frontier-lab client under NDA)

AI Agent Evaluator · Remote · 2026

- Designed realistic engineering tasks against production-grade codebases to elicit and capture reproducible behavioural failures in an autonomous coding agent.
- Authored grader guidance: privileged rubrics consumed by an automated LLM grader, calibrating scoring across seven behavioural dimensions.
- Produced 4+ reference runs per task to prove failure reproducibility and validate that rubrics discriminate strong from weak agent behaviour.
- Iterated rubrics through regrade cycles, re-scoring frozen agent trajectories to isolate the effect of each rubric change on the score distribution.
- Applied an ~80%-of-senior-engineers consensus bar to separate material failures with real business impact from stylistic preference.
- Contributed within a large distributed evaluation programme; joined recurring client syncs on task design and grading standards.

Stack: LLM evaluation, RLHF data generation, rubric design, Docker-based agent sandboxes, Ruby on Rails, TypeScript

HomeMoney — Multi-Tenant Financial Analytics

Full-Stack Developer · United States

- Developed an enterprise multi-tenant financial ledger and analytics platform centralising accounting, cash-flow management and financial reporting across complex organisational structures.
- Architected backend services for ledger management, accounts receivable and payable, asset reconciliation and configurable financial calculations, with granular role-based access control and secure data isolation for individual companies and business groups.
- Built automated background pipelines that import and normalise financial data from banking systems and external sources, enabling high-volume transaction processing and reconciliation.
- Developed a dynamic formula engine that lets administrators configure custom financial calculations and reporting logic without code changes.
- Implemented financial dashboards, balance-sheet reporting, cash-flow analytics and secure data export, applying AES-256 encryption and key management to protect sensitive data.

Stack: TypeScript, Node.js, Express.js, PostgreSQL, TypeORM, Redis, BullMQ, MinIO, Elasticsearch, Docker

Inkasso — Enterprise Debt Collection & Financial Automation

Full-Stack Developer · Switzerland, Germany, Italy

- Developed and enhanced an enterprise multi-tenant debt-collection and financial automation platform serving organisations across Italy, Germany and Switzerland.

- Architected backend services automating debtor lifecycle management, invoicing, payment reconciliation, interest and legal-cost calculations and repayment schedules, while supporting tenant-specific business rules and country-specific compliance requirements.
- Designed asynchronous processing pipelines on Redis and BullMQ to orchestrate high-volume financial operations: automated data ingestion, payment imports, scheduled reconciliations and document processing.
- Built dynamic document generation for invoices, legal notices, payoff statements, reports and Swiss QR-bills, and integrated email and SMS communication channels.
- Implemented bulk reporting, document management and tenant synchronisation, and optimised the platform for reliability through background workers, scheduled tasks and robust transactional workflows.

Stack: TypeScript, NestJS, Node.js, PostgreSQL, TypeORM, Redis, BullMQ, RabbitMQ, Swagger/OpenAPI, Jest, Docker

Longevity — AI-Driven Healthcare Platform

Full-Stack Developer · Latvia

- Built a clinical genetic and blood-analysis platform on NestJS and the OpenAI API that generates detailed medical LLM assessments, historical health trends and marker-specific recommendations.
- Architected an event-driven system that reacts to a completed AI analysis by instantly triggering patient notifications, updating doctor dashboards and automating post-analysis workflows.
- Delivered a highly responsive SPA with React, Vite, Tailwind CSS and TanStack Query, using Zustand for complex state across patient portals and comprehensive admin dashboards.
- Built a stateless REST API on Node.js, PostgreSQL and TypeORM, managing secure AWS S3 file storage, automated OpenAPI client generation and Paysera payment integration for consultation bookings.

Stack: TypeScript, React, Vite, NestJS, Node.js, PostgreSQL, TypeORM, Tailwind CSS, TanStack Query, Zustand, AWS S3, OpenAI API, Paysera

EyeVi — LIDAR Mapping System

Full-Stack Developer · Estonia

- Built a scalable platform for high-accuracy 3D mapping and digital road-infrastructure inventory.
- Engineered a real-time visualisation system for 3D-scanned streets, using Three.js and Potree to render massive LIDAR point clouds directly in the browser.
- Architected a NestJS backend on PostgreSQL with the PostGIS extension to calculate complex geospatial statistics, and integrated ArcGIS for enterprise spatial analytics.
- Engineered a token-based subscription architecture on Stripe to manage automated billing and API usage quotas.
- Implemented high-volume geodata upload workflows with AWS S3 and optimised rendering performance through Redis caching.

Stack: TypeScript, React, Three.js, Potree, Mapbox, ArcGIS, NestJS, PostgreSQL, PostGIS, Redis, AWS S3, Stripe, Docker

Rozumnyk — AI-Powered Agriculture Knowledge Assistant

Full-Stack Developer · Ukraine

- Designed and delivered an enterprise-grade AI assistant platform focused on high-quality Ukrainian-language interactions, combining modern LLMs with a Retrieval-Augmented Generation (RAG) pipeline.
- Architected a modular NestJS backend responsible for document ingestion, asynchronous processing, vector embedding generation and semantic retrieval, using Redis, BullMQ, Ollama and pgvector for scalability and low-latency responses.
- Established a role-based authorisation model for administrators, moderators and end users, and developed operational dashboards for content and system management.

- Containerised the complete solution with Docker and Kubernetes, with MinIO providing secure object storage for knowledge assets.

Stack: TypeScript, NestJS, Next.js, PostgreSQL, pgvector, Redis, BullMQ, Vercel AI SDK, Groq API, Ollama, MinIO, Docker, Kubernetes, Material UI

Baulera — Multi-Tenant Invoicing & Accounting SaaS

Full-Stack Developer · Lithuania

- Led the architecture and full-stack development of a multi-tenant financial SaaS platform centralising invoicing, procurement and accounting workflows across multilingual business environments.
- Designed scalable backend systems with Express, MySQL and Sequelize, implementing an RBAC architecture for secure company hierarchies and delegated workflows.
- Built automation-driven financial pipelines using Redis and scheduled jobs for recurring billing, payment tracking and communication workflows.
- Integrated Stripe payment infrastructure, developed document automation for receipt processing and PDF generation, and delivered React financial dashboards with the Finvalda, Ivesk.lt and Rekvizitai.lt accounting integrations.
- Implemented enterprise-grade authentication, GDPR-compliant data lifecycle management and secure multi-company data handling.

Stack: TypeScript, React, Redux Saga, Node.js, Express.js, MySQL, Sequelize, Redis, Stripe, Socket.IO, Puppeteer, Material UI, AWS S3

ASPRA — Learning Management System

Full-Stack Developer · Ukraine

- Developed a scalable learning management system on Next.js and NestJS, enabling dynamic course creation, integration of complex SCORM modules and advanced assessment workflows.
- Created a secure authentication architecture providing strict role-based access control across comprehensive control panels for administrators, instructors and students.
- Implemented an interactive task management engine handling automated testing, multimedia assignment submissions and structured instructor grading, plus real-time chat infrastructure and precise user progress tracking.
- Optimised backend REST APIs on PostgreSQL and TypeORM, and engineered a globally distributed media delivery pipeline with AWS S3 and CloudFront for seamless video streaming.
- Streamlined deployment and reliability by establishing Docker containerisation and automated CI/CD pipelines.

Stack: TypeScript, Next.js, NestJS, PostgreSQL, TypeORM, WebSockets, Redis, AWS S3, AWS CloudFront, Docker, CI/CD

Saveface — Face-Value Ticket Exchange

Full-Stack Mobile Developer · United States

- Built a cross-platform mobile application with a scalable backend for face-value ticket exchange, developing the native client in React Native and Expo.
- Engineered a robust, Swagger-documented REST API with NestJS and PostgreSQL via TypeORM to enforce complex marketplace logic, including secure 72-hour payout holds and fraud review workflows.
- Coordinated verified digital ticket transfers through Ticketmaster and AXS.
- Built high-concurrency automated auto-purchase queue systems and general-admission waitlists on Redis and BullMQ, with Stripe payments, Sentry monitoring and Helmet security hardening.

Stack: TypeScript, React Native (Expo), NestJS, Node.js, PostgreSQL, TypeORM, Redis, BullMQ, Stripe, Sentry, Helmet

Fast Forward — Transportation Management System

Full-Stack Developer · United States

- Developed a scalable multi-tenant TMS for freight brokerage operations, supporting the complete shipment lifecycle from quoting and load creation to dispatching, tracking and invoicing.
- Built responsive brokerage dashboards for dispatchers, carrier managers and customers with secure role-based access control and tenant isolation.
- Implemented carrier onboarding workflows, rate confirmation management, document processing and automated invoicing.
- Engineered asynchronous background processing with BullMQ and Redis for EDI imports, tracking updates, invoice generation and document synchronisation.
- Designed real-time shipment monitoring over WebSockets and deployed containerised services on AWS with automated CI/CD pipelines.

Stack: TypeScript, React, Next.js, NestJS, PostgreSQL, TypeORM, Redis, BullMQ, WebSockets, AWS (S3, ECS, RDS), Docker, CI/CD

WelSpot — Medical Management System

Full-Stack Developer · United States

- Engineered a scalable medical management platform on React and NestJS to streamline complex appointment scheduling and patient and staff profile management.
- Enabled real-time application state synchronisation with Socket.IO and RabbitMQ messaging.
- Developed and documented robust REST APIs via Swagger on PostgreSQL and Prisma, and implemented Elasticsearch to deliver rapid fuzzy search across millions of clinical notes and medical records.
- Enforced strict JWT-based role-based access control and compliant document workflows, using AWS S3 for secure medical file storage.

Stack: TypeScript, React, NestJS, PostgreSQL, Prisma, Redis, RabbitMQ, Elasticsearch, Socket.IO, AWS S3, Swagger